```
# Why do we need Window Function
select * from employees.employees; # 107 rows

# get me total salary that a company xyz pays?
SELECT select sum(salary) from employees.employees e; # 1 row
# My data is getting shrunk into a single group for me to get the salary


# get me total salary that a company xyz pays at every dept level?
select d.department_name , sum(salary) as dept_salary
from employees.employees e
left join employees.departments d
on e.department_id = d.department_id
group by d.department_name;
# 12 rows -> 12 unique department IDS that we have in our table
# The data is getting shrunk from 107 records to 12 records


# show me total salary across every employees?
# Part 1 - Using SubQuery  # 6,91,400.00 all 107 records
select
	first_name,
	employee_id,
	salary,
	department_id,
	(select sum(salary) from employees.employees) as tot_salary_for_all_emp
from employees.employees e;
# Part 2 - Using Window Function
select
	first_name,
	employee_id,
	salary,
	department_id,
	sum(salary) over() as tot_salary_for_all_emp
from employees.employees e;
```

```
# Unknown column 'tot_salary_for_all_emp' in 'where clause'
# farmers_market
# avg sales value per customer per quantity in separate column
select
	*,
	round(
		avg(cost_to_customer_per_qty) over(), 2
	) as avg_cost_to_customer_per_qty
from farmers_market.customer_purchases cp ;


# get me average salary that a company xyz pays at every dept level?
select
	e.department_id,
	round(avg(salary), 2) as avg_salary_dept
from employees.employees e
group by department_id
order by 2 desc;
# 107 records -> 12 records



# get me total salary that a company xyz pays at every dept level displayed at every
row?
select
	first_name, -- 1
	employee_id, -- 2
	salary, -- 3
	department_id, -- 4
	round(
		avg(salary) over(partition by department_id), 2
	) as avg_salary_dept -- 5
from employees.employees
order by 5 desc;
-- how many times will I scan through my records in this table?
-- there are 12 departments
-- however we will only scan the table once
# got all 107 records here
```

```sql
# get row number for each of the employees sorted on desc order based on salary
select
	first_name, -- 1
	employee_id, -- 2
	salary, -- 3
	department_id, -- 4
	row_number() over(order by salary desc) as r_no
from employees.employees;
# get me 13th highest salary from employees -> very frequenty asked in interview
select * from (
	select
		first_name,
		employee_id,
		salary,
		department_id,
		row_number() over(order by salary desc) as r_no,
		rank() over(order by salary desc) as rnk,
		dense_rank() over(order by salary desc) as dense_rnk
	from employees.employees
) t
where t.dense_rnk = 13;
-- row number does not work for duplicate values
-- this is not the correct approach due to inability to handle duplicate values
select distinct salary from employees.employees e
order by salary desc; -- 9000 is the 13th highest salary
# get me avg dept_wise_salary across every row sorted by first name
select
	first_name,
	employee_id,
	salary,
	department_id,
	round(
		sum(salary) over(partition by department_id order by first_name), 2
	) as avg_salary_per_dept
from employees.employees
where department_id = 90;

# get me the sum of salary for employees ordered by joining date
```